

HARSHA VARDHAN BALEPALLI

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Education

National Institute of Technology, Rourkela

Bachelor of Technology in Electronics and Instrumentation Engineering (CGPA: 8.26)

Aug. 2024 – May 2028

Rourkela, Odisha

Sri Chaitanya Institutes

Intermediate (93%)

May 2023

Vijayawada, Andhra Pradesh

Projects

Full-Stack AI Assistant | *React, Flask, LangChain, Groq (Llama 3.3), Python*

- Built a full-stack AI assistant using React and Flask, integrating LangChain with Llama 3.3 via Groq for conversational interactions.
- Implemented tool calling for calculator, weather, and movie recommendation services using OpenWeatherMap and TMDB APIs.
- Designed session-based conversation memory and multi-turn chat handling using LangChain message abstractions and configurable history management.
- Developed a modular backend architecture with separated LLM, tool, and API layers, and built REST APIs with CORS support and robust error handling.

Movie Discovery Web Application | *React, Appwrite, TMDB API*

- Built a responsive movie discovery platform using React and TMDB REST APIs for dynamic content rendering.
- Implemented search optimization with custom debouncing to reduce unnecessary API calls and improve application performance.
- Integrated Appwrite backend services to store and display real-time trending search data.

Full-Stack Blog Platform | *Node.js, Express.js, MongoDB, EJS, JWT*

- Built a full-stack blog platform enabling users to create, edit, delete and manage blog content.
- Implemented JWT-based authentication and protected routes using secure cookie handling for user access control.
- Designed a comment system connecting users and blogs to support interactive content engagement.
- Integrated password security using salt and HMAC SHA-256 hashing along with image upload functionality using Multer.

Leadership & Community Experience

Google Campus Ambassador

March 2026 – Present

- Selected to represent Google technologies at NIT Rourkela and facilitate technical engagement within the student community.

Achievements / Certifications

- **Data Structures & Algorithms:** Solved 400+ DSA problems across coding platforms with emphasis on dynamic programming, graphs, recursion, and problem solving.
- **Reliance Foundation Scholar:** Awarded scholarship for exceptional performance in a competitive nationwide assessment.

Technical Skills

- **Languages:** Java, C, JavaScript, HTML/CSS
- **Technologies/Frameworks:** React.js, Node.js, Express.js, Flask, LangChain, MongoDB, Appwrite, Bootstrap, Python
- **Developer Tools:** Git, GitHub, VS Code, REST APIs, Groq
- **Core Subjects:** Data Structures, Algorithms, Object Oriented Programming and Design, Database Management

Extracurricular Activities

- Active member of Open Code, driving **collaborative** problem-solving and peer learning within a **fast-paced**, results-driven campus developer community.